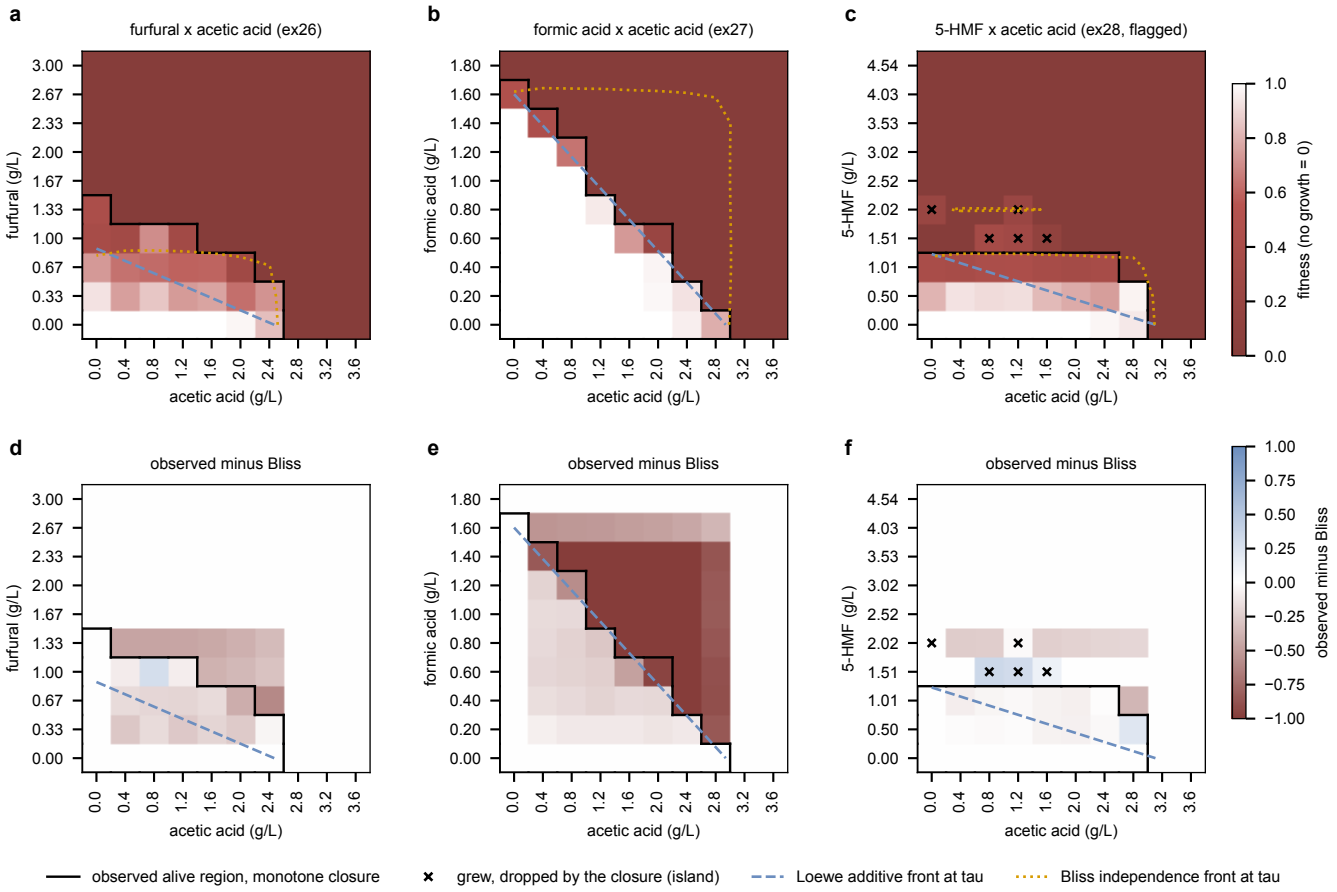


Figure 11: No single rule draws the boundary. **a** to **c**, Observed fitness over each isobole grid (no growth drawn as zero), the black outline around the monotone closure of the cells that grew, a cross on each grown cell the closure drops, and the two fronts at the grid's growth threshold: Loewe additivity dashed in blue, Bliss independence dotted in orange. Furfural with acetic acid follows Bliss, overlap 0.733 against 0.286 for Loewe, which calls 10 grown cells dead. Formic with acetic acid follows Loewe, overlap 0.875 against 0.436 for Bliss, which keeps 31 dead cells alive. 5-HMF with acetic acid follows Bliss, overlap 0.765 against 0.385. The growth edge along acetic acid tracks the winning rule within 0.444 (furfural), 0.333 (formic acid) and 0.778 (5-HMF) steps on average. ✗● **d** to **f**, Observed minus the empirical Bliss surface on a symmetric scale, with the Loewe front dashed. ✗●